Attention Isn't All You Need

Aircraft Classification from Climb Trajectories

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1 Abstract

Identifying aircraft type from ADS-B trajectory data during the initial climb phase has applications in airspace monitoring, performance modelling, and air traffic management. This work frames the problem as a multivariate time-series classification task: given the first 1,200 seconds of a flight's climb, predict which of 11 commercial aircraft types generated it. We evaluate four approaches: a Nearest Mean Curve baseline, a 1D CNN, a Temporal Convolutional Network (TCN), and a Transformer encoder, across both a balanced sample dataset and a large-scale imbalanced full dataset. The TCN achieves the best overall performance, reaching 69.1 % accuracy on the 550-segment sample and 81.8 % accuracy (macro F1 = 0.755) on 1.33 M test segments from the full dataset. The Transformer underperforms the TCN, suggesting that local temporal structure matters more than global attention for short-horizon climb profiles. Narrowbody variants (A319/A320/A321, E190/E195) remain the principal source of confusion, while aerodynamically distinct types (DH8D, B77W) are reliably classified across all models.

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2 Introduction

Modern air traffic surveillance generates continuous ADS-B position reports for every tracked aircraft. Each message carries barometric altitude, groundspeed, vertical rate, and a 24-bit ICAO identifier. Reliable aircraft type information often depends on secondary databases (e.g., operator fleet records), which can be incomplete or delayed. Automated inference of aircraft type from raw trajectory data would remove this dependency and enable downstream applications such as fuel-burn estimation, wake-turbulence categorisation, and anomaly detection.

The climb phase is a natural window for this task. Manufacturers design takeoff and initial climb procedures to specific performance envelopes, so each type leaves a characteristic fingerprint in altitude gain rate, airspeed build-up, and vertical speed profile. At the same time, families of aircraft (e.g., the A320 family) share aerodynamic heritage, making fine-grained discrimination difficult.

This project investigates whether short time-series classifiers, from a nearest-mean baseline to deep sequence models, can reliably distinguish 11 commercial aircraft types using only three observable climb-phase signals.

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3 Related Work

The closest prior work on this dataset is Alligier and Gianazza (2018), who assembled the same OpenSky 2017 climbing segments and used machine learning to predict the unknown point-mass model parameters (aircraft mass and speed intent) needed for ground-based trajectory prediction. Training on the first ten months and testing on the last two, their ML approach reduced altitude RMSE by 48 % and speed RMSE by 25 % relative to a BADA-with-mean-parameters baseline, with further gains when pre-takeoff information was used to predict parameters ahead of time. That work treats the trajectory as a regression target; the present study reframes the same data as a classification problem, asking which aircraft type generated a given climb rather than what it will do next. To our knowledge, direct aircraft type identification from ADS-B climb trajectories has received less attention than trajectory prediction, making this a natural extension of Alligier and Gianazza's dataset to a supervised recognition task.

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4 Data and Preprocessing

4.1 Source Data and Features

The dataset is the OpenSky Network Climbing Aircraft Dataset (Alligier 2017), curated by Richard Alligier (ENAC) from ADS-B traffic recorded throughout 2017. It contains millions of climbing segments from aircraft worldwide, covering the 11 most frequent commercial types: A319, A320, A321, A332, B737, B738, B77W, CRJ9, DH8D, E190, E195. The dataset was originally assembled to train machine learning models predicting missing performance parameters (altitude and airspeed)

on a 10-minute horizon, where an ML approach reduced altitude RMSE by 48 % and speed RMSE by 25 % relative to a BADA baseline. Here we repurpose the same trajectories for aircraft type classification.

Three features are retained per timestep:

Feature	Unit	Description
`baroaltitude`	m	Barometric altitude (primary climb signal)
`vertratecorr`	m/s	Corrected vertical rate
`tas`	m/s	True airspeed

Each segment is truncated to $t \in [0, 1200]$ s from departure. All features are standardised with a `StandardScaler` fit on training data only, then zero-padded to the maximum observed sequence length.

4.2 Dataset Splits

Sample dataset (balanced, used for baseline and initial model comparisons):

- Train: 500 randomly sampled segments per type (5,500 total)
- Test: 50 segments per type (550 total)

Full dataset:

Split	Segments
Train	2,769,261
Valid	856,670
Test	1,325,101

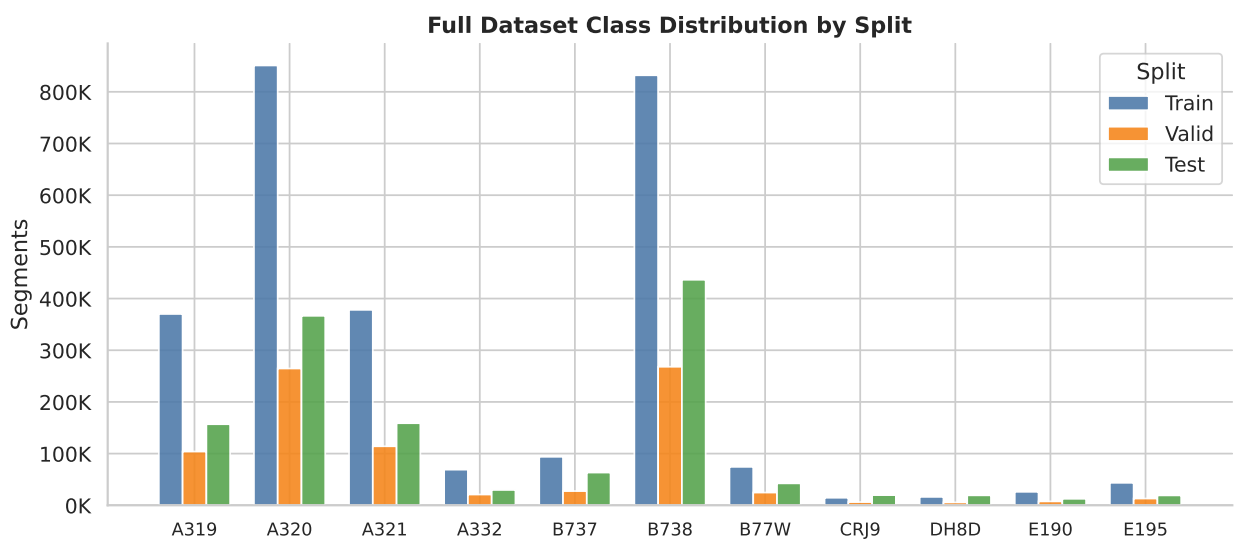

Figure 1. Full dataset class distribution. Segment counts per aircraft type and split. A320 and B738 together account for ~60 % of training data; DH8D and CRJ9 are the rarest classes, motivating a weighted sampler during training.

4.3 Average Climb Profiles

Figure 2 shows the mean flight level, true airspeed, and vertical speed for each aircraft type, averaged over the 500 training segments per class. Aerodynamically distinct types (DH8D, B77W) occupy clearly separate regions, while the A319/A320/A321 family and the E190/E195 pair are visually near-identical across all three panels.

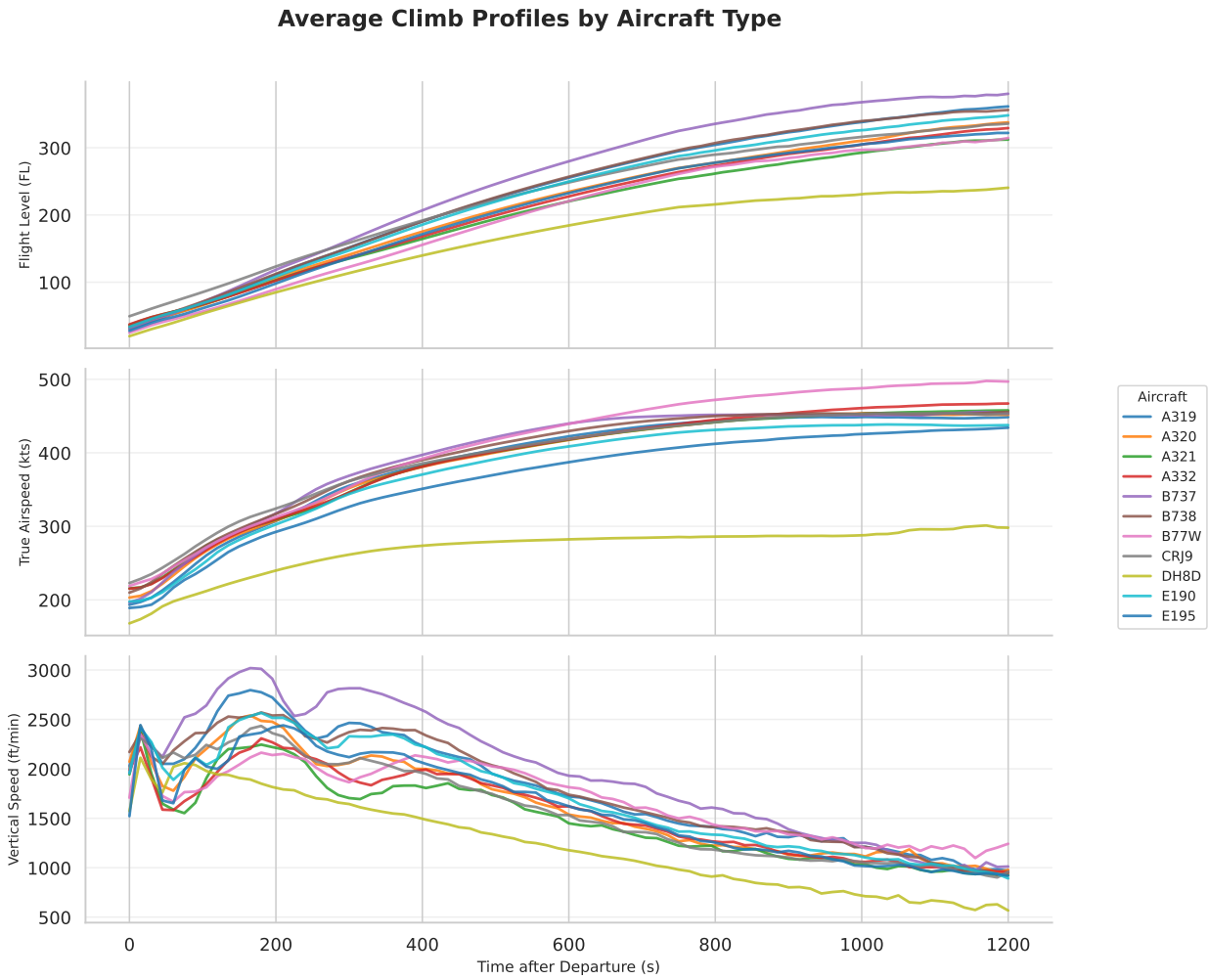

Figure 2. Mean climb profiles by aircraft type. Three panels show flight level, true airspeed, and vertical speed averaged over 500 training segments per type. DH8D climbs slowly to low FL; B77W climbs steeply to high altitude. A319/A320/A321 traces largely overlap, foreshadowing their mutual confusion.

5 Methods

5.1 Sample Dataset Experiments

Three models were developed and evaluated on the balanced 500/50-segment-per-class sample dataset.

5.1.1 Nearest Mean Curve (Baseline)

For each class, compute a prototype curve by averaging all training segments at each timestep. At test time, assign the label of the nearest prototype under L2 distance over shared timesteps:

$$\hat{c} = \arg\min_{c} \sqrt{\sum_{t \in \mathcal{T}_s \cap \mathcal{T}_c} \|\mathbf{s}(t) - \bar{\mathbf{x}}_c(t)\|^2}$$

No learned parameters; purely an interpretability lower bound.

5.1.2 1D Convolutional Neural Network (CNN)

Three-block 1D CNN on tensors of shape `(batch, features, timesteps)`:

Block	Channels	Kernel	Pooling
1	$3 \to 64$	7	MaxPool(2)
2	$64 \to 128$	5	MaxPool(2)
3	$128 \to 256$	3	AdaptiveAvgPool(1)

Classifier: `Dropout(0.3) → Linear(256→128) → ReLU → Dropout(0.2) → Linear(128→11)`. Trained for 50 epochs, Adam lr=1e-3, `ReduceLROnPlateau` (patience=5, factor=0.5).

5.1.3 Temporal Convolutional Network (TCN)

Three `TCNBlock` layers with residual connections and exponentially growing dilation:

Block	Channels	Dilation
1	$3 \to 64$	1
2	$64 \to 128$	2
3	$128 \to 256$	4

Kernel size 3; dropout 0.2 within blocks, 0.3 pre-classifier. Dilated convolutions allow the network to observe the entire climb trajectory without deep stacking. Global average pooling collapses the temporal axis before a two-layer MLP (256→128→11). Trained for 50 epochs, Adam lr=1e-3, batch 32.

5.2 Full Dataset Experiments

The TCN and a Transformer encoder were then trained on the full 2.77 M segment dataset, with a `WeightedRandomSampler` to counter class imbalance and mixed-precision AMP throughout.

5.2.1 TCN (Full Dataset)

Same architecture as above. Adam lr=1e-3, batch 512, best checkpoint selected on validation accuracy.

5.2.2 Transformer Encoder

CLS-token Transformer encoder:

- Linear projection: 3 → d_model=64 per timestep; learnable positional embeddings
- 4 Pre-LN encoder layers, 4 heads, d_ff=256
- CLS token → Dropout → Linear(64→11)

Training: AdamW lr=3e-4, OneCycleLR (5-epoch warmup over 50 epochs), gradient clipping, AMP.

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6 Results

6.1 Sample Dataset

6.1.1 Nearest Mean Curve (Baseline)

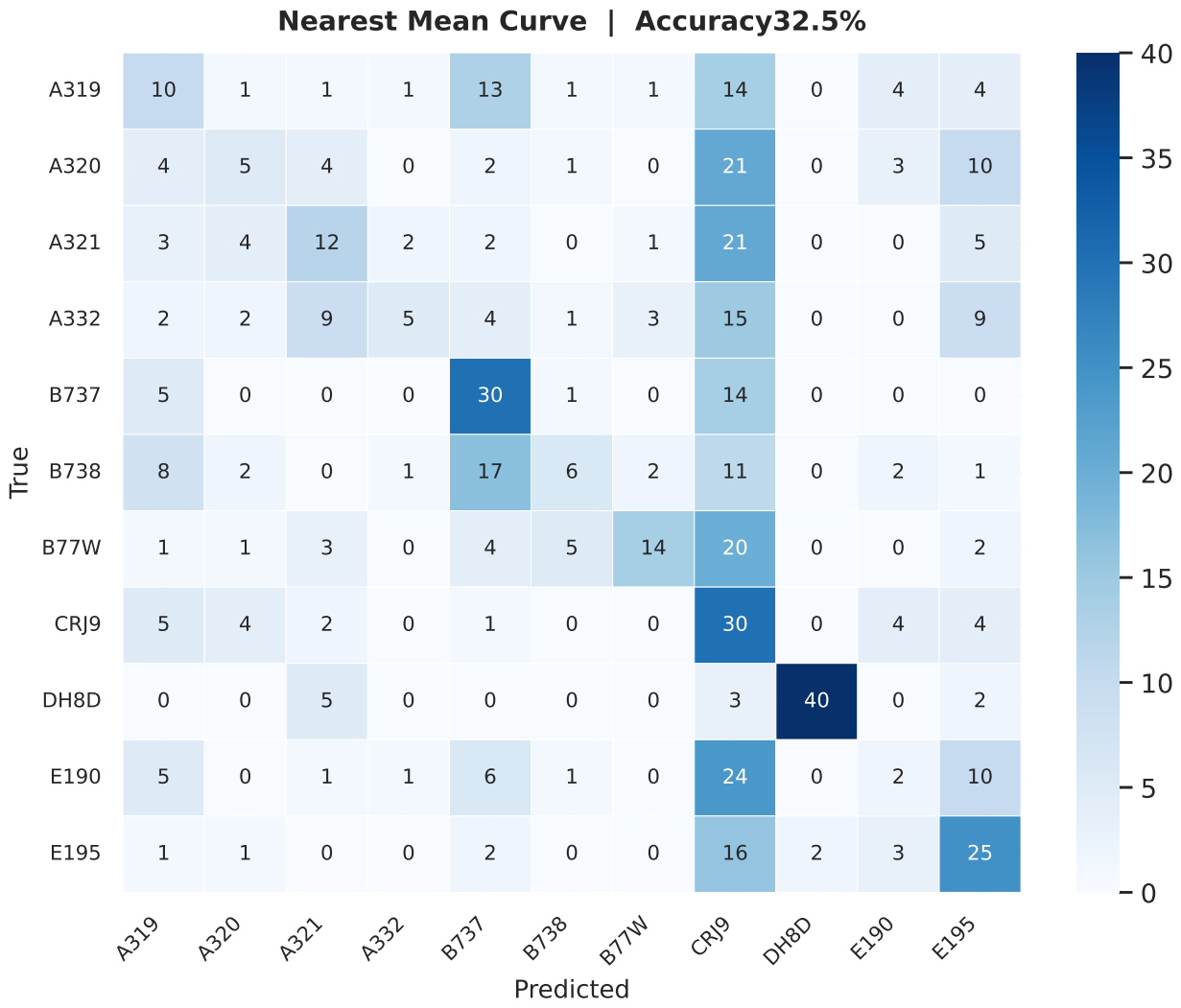


Figure 3. Nearest Mean Curve confusion matrix (sample dataset, n=550). DH8D and B77W separate cleanly; the A319/A320/A321 block shows heavy mutual confusion, as does the E190/E195 pair.

Overall accuracy: 32.5%. Random chance on 11 balanced classes is 1/11 (or approx. 9.1%), so even this parameter-free baseline is well above floor. The classifier already separates the turboprop (DH8D) and widebody (B77W, A332) archetypes, but the Airbus narrowbody family and the Embraer regional pair collapse into each other.

6.1.2 1D CNN

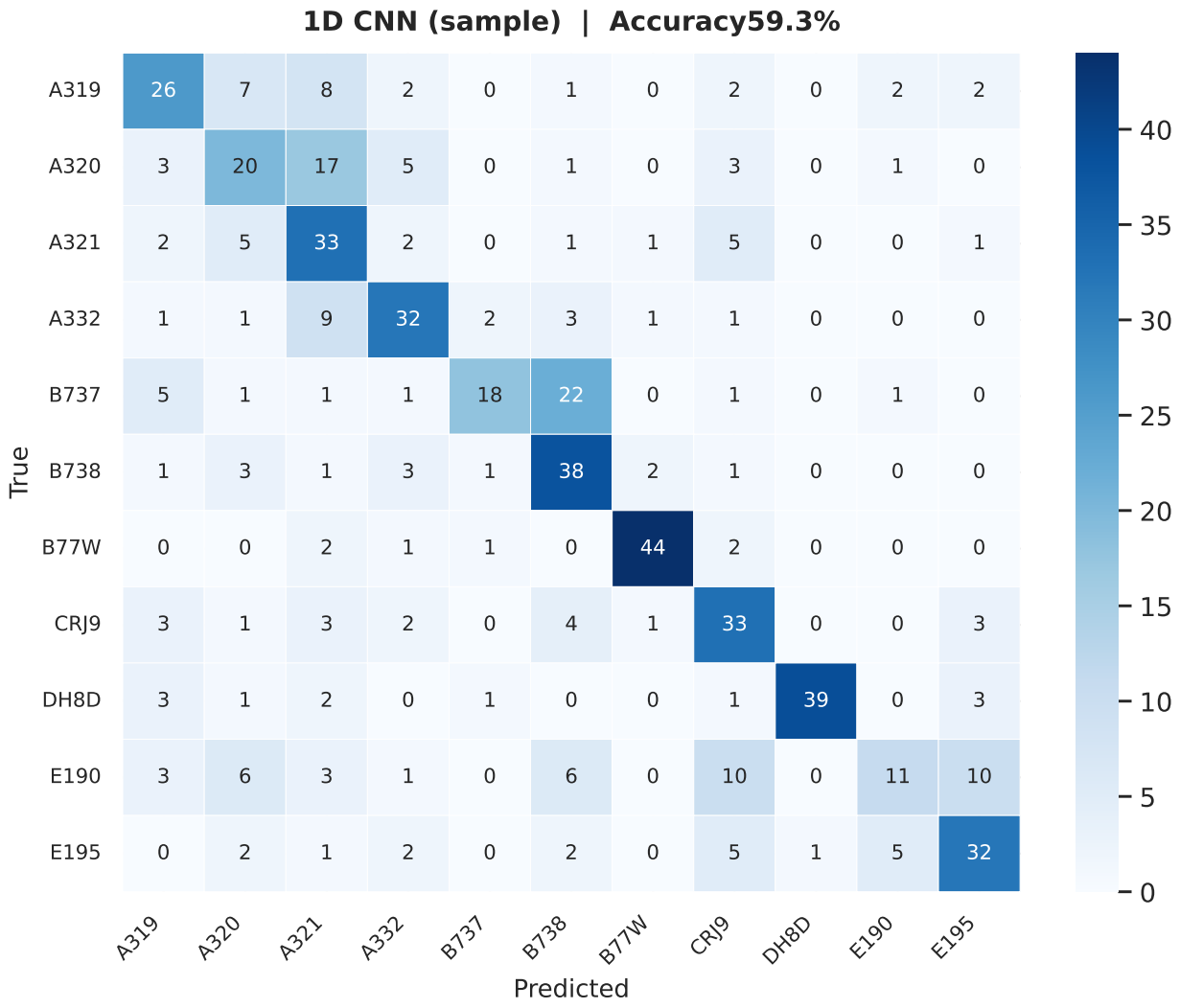


Figure 4. 1D CNN confusion matrix (sample dataset, n=550). E190 recall collapses largely absorbed by A319 and E195. B77W and DH8D remain well-separated.

Table 5: Per-class metrics, 1D CNN (sample dataset)

Aircraft	precision	recall	f1-score	support
A319	0.553	0.52	0.536	50
A320	0.426	0.40	0.412	50
A321	0.412	0.66	0.508	50
A332	0.627	0.64	0.634	50
B737	0.783	0.36	0.493	50
B738	0.487	0.76	0.594	50
B77W	0.898	0.88	0.889	50
CRJ9	0.516	0.66	0.579	50

Table 5: Per-class metrics, 1D CNN (sample dataset)

Aircraft	precision	recall	f1-score	support
DH8D	0.975	0.78	0.867	50
E190	0.550	0.22	0.314	50
E195	0.627	0.64	0.634	50

6.1.3 TCN

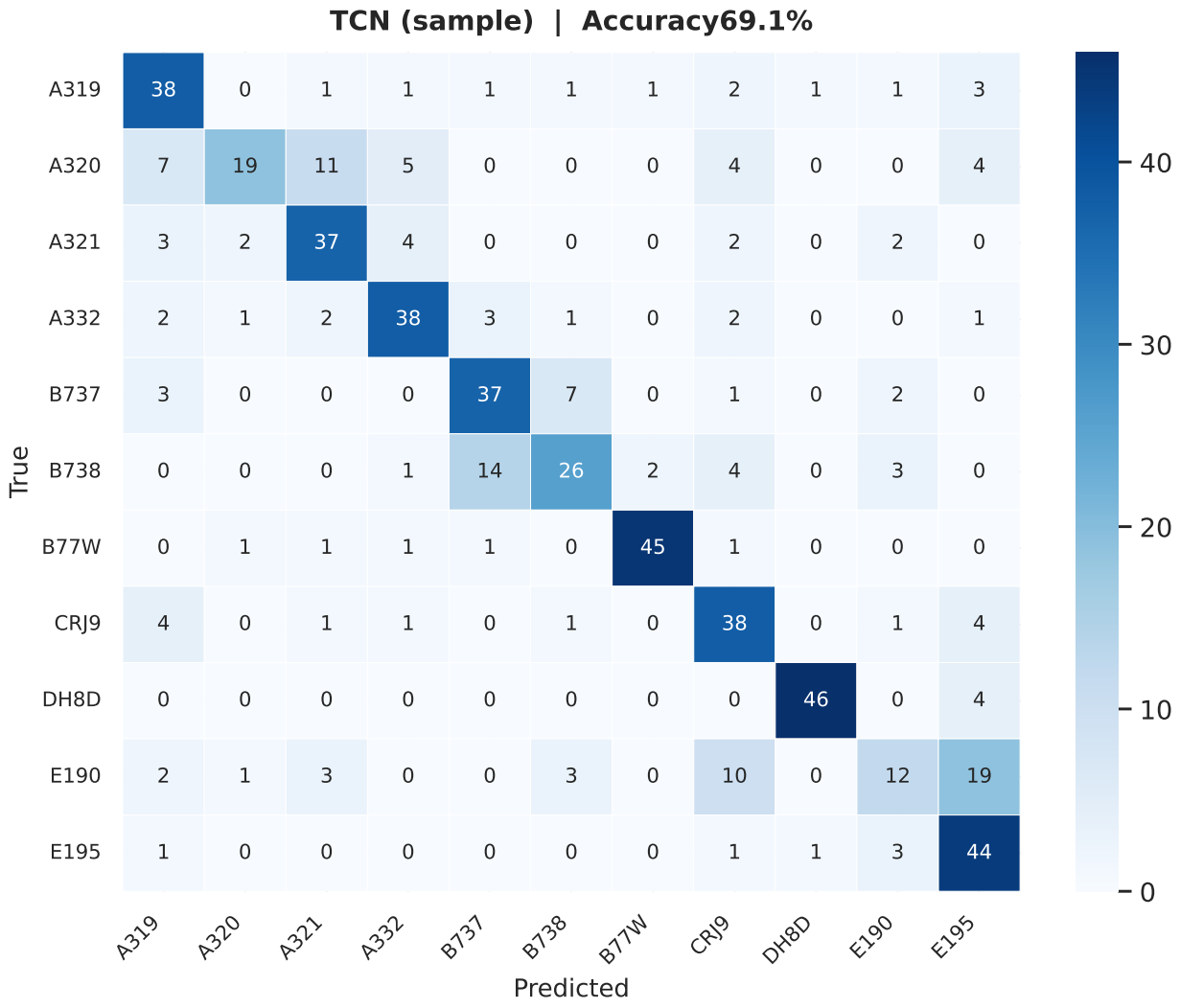


Figure 5. TCN confusion matrix (sample dataset, n=550). Dilated receptive fields better capture the full arc of the 1,200-second climb, yielding a substantial improvement over the CNN.

6.2 Full Dataset

6.2.1 TCN

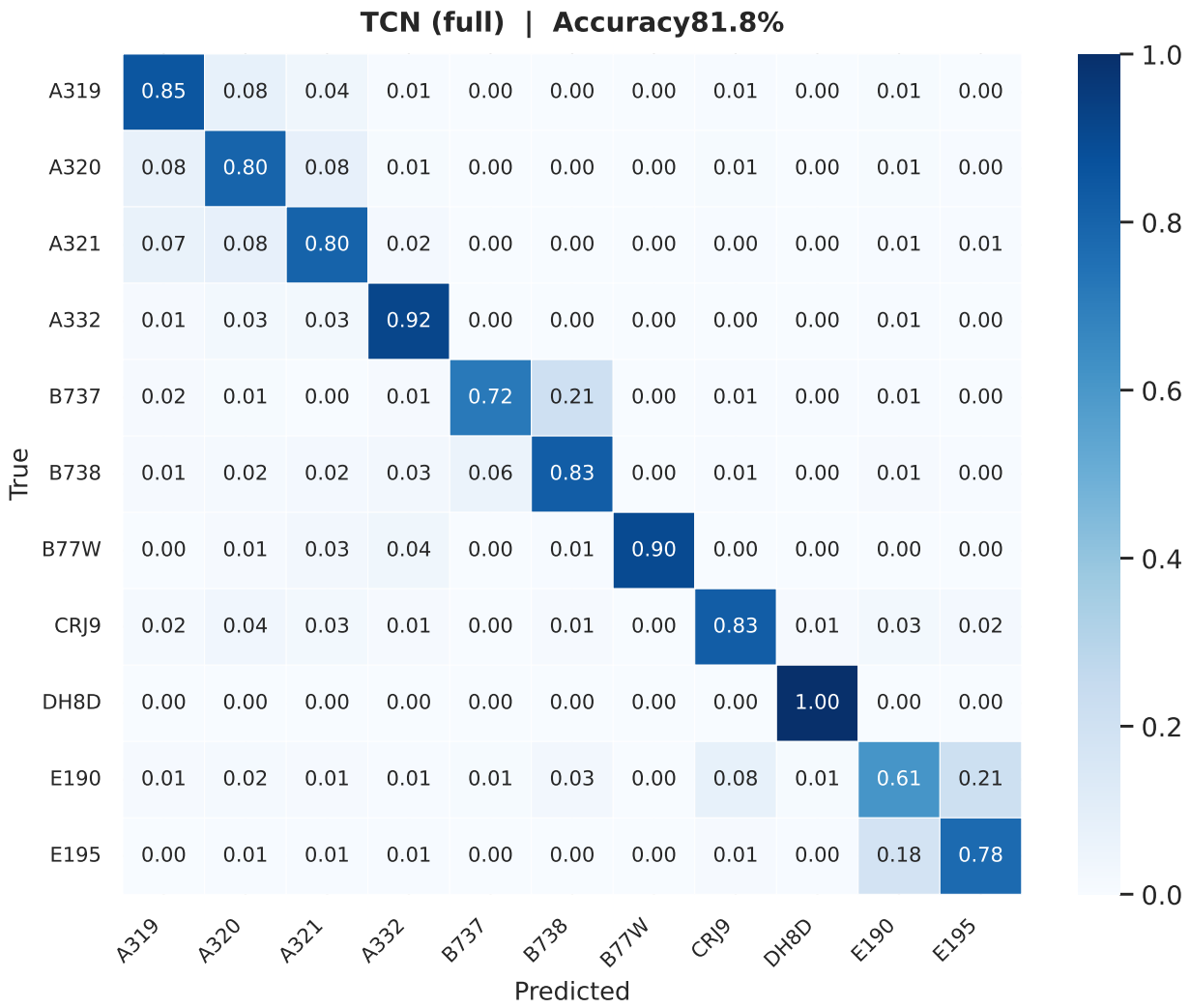


Figure 6. TCN confusion matrix (full dataset, n=1,325,101, row-normalised). DH8D achieves near-perfect recall (99.9 %). A332 precision is low, indicating the model over-predicts widebody status. E190 remains the weakest class.

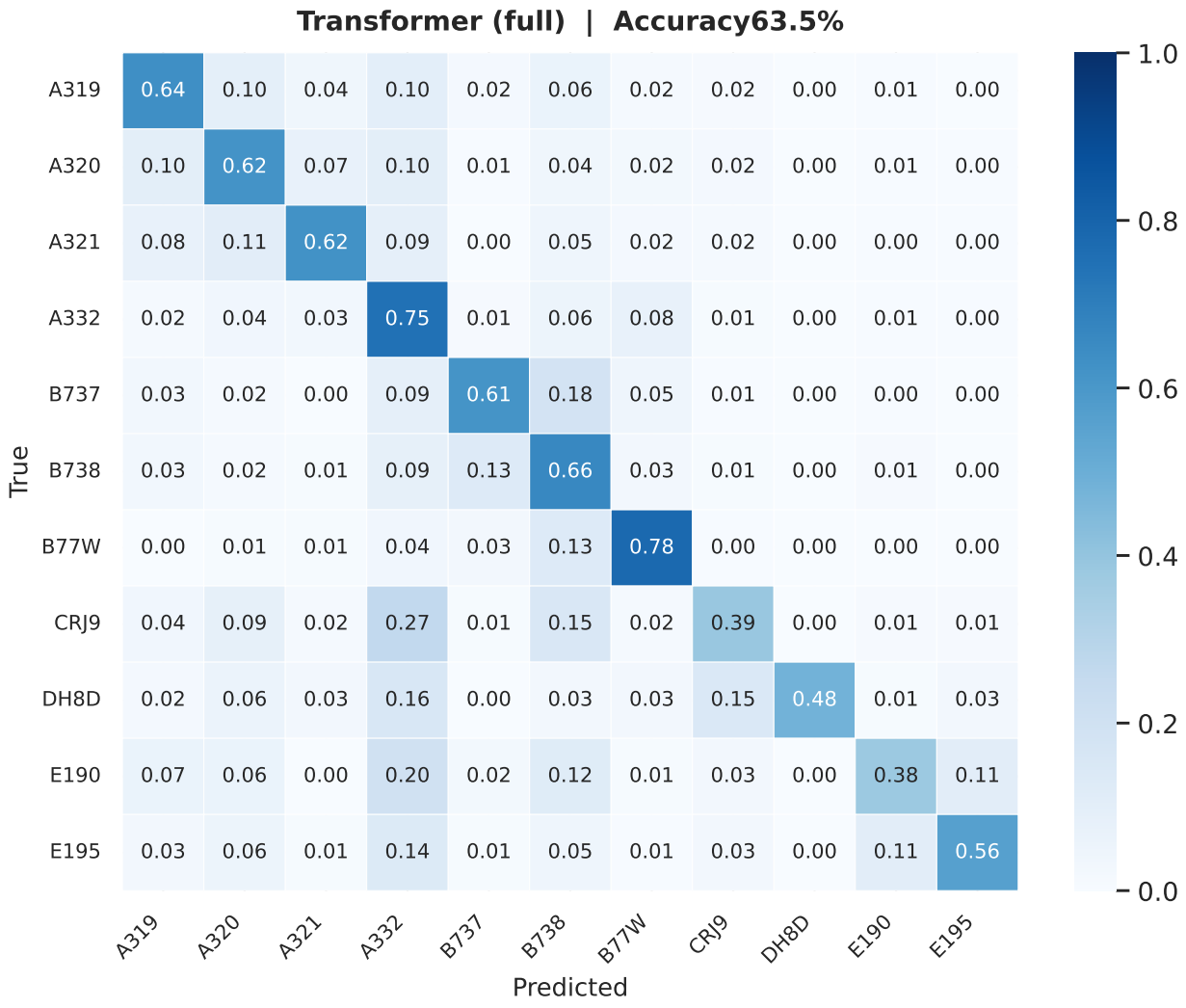
Table 6: Per-class metrics, TCN (full dataset)

Aircraft	precision	recall	f1-score	support
A319	0.736	0.853	0.790	156897
A320	0.888	0.800	0.842	366647
A321	0.722	0.801	0.759	158842
A332	0.535	0.915	0.675	29712
B737	0.624	0.721	0.669	63383

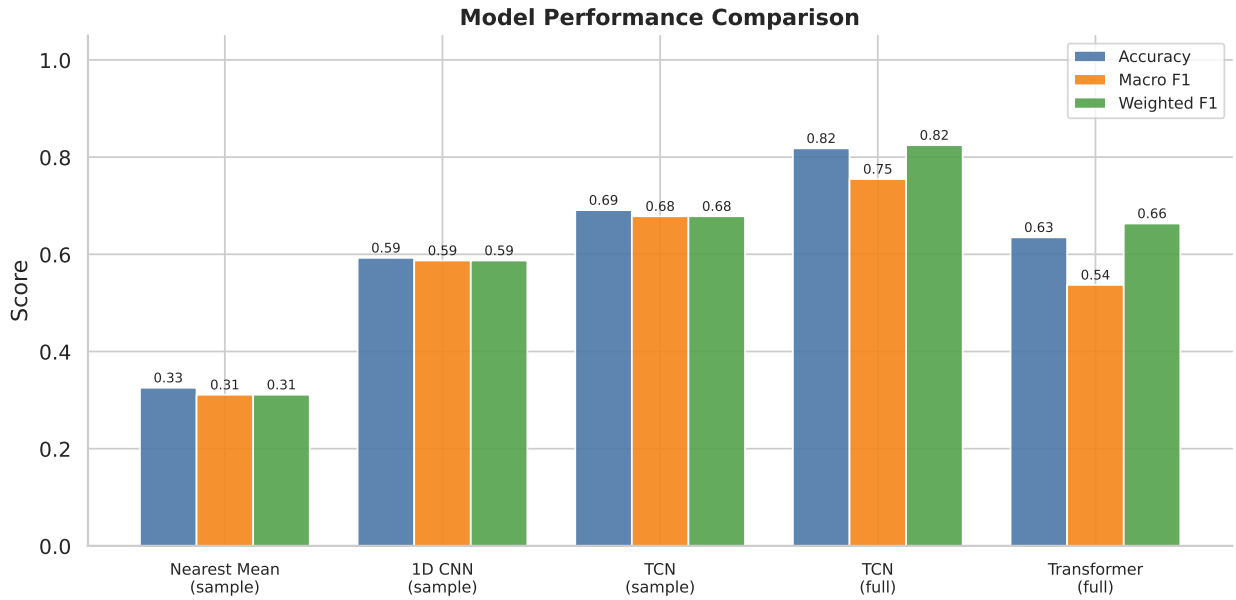
Table 6: Per-class metrics, TCN (full dataset)

Aircraft	precision	recall	f1-score	support
B738	0.958	0.826	0.887	436404
B77W	0.944	0.899	0.921	42468
CRJ9	0.626	0.825	0.712	19699
DH8D	0.882	0.999	0.937	19290
E190	0.312	0.610	0.412	12589
E195	0.635	0.777	0.699	19170

6.2.2 Transformer



6.3 Overall Comparison



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7 Discussion

7.1 Class Confusion Patterns

The A319/A320/A321 triplet shares engines, flight management systems, and operator climb procedures; their altitude, airspeed, and vertical rate profiles are nearly indistinguishable in the first 20 minutes post-takeoff.

E190 is persistently the hardest class (F1 = 0.32-0.41 across all models). The Embraer 190/195 are closely related, and the E190's performance envelope overlaps with lighter narrow-bodies at similar load factors.

B737 vs B738: B738 is well-classified (F1 = 0.887, full TCN) but the generic B737 label scores only 0.669, likely because it aggregates multiple heterogeneous 737 variants.

DH8D (a turboprop) stands apart; all models identify it reliably, with recall approaching 100 % on the full TCN.

7.2 Data Scale vs Architecture

Change	Accuracy gain
CNN to TCN (sample data)	+9.8 pp
TCN sample to TCN full	+12.7 pp

Data scale wins. The model was severely data-limited at sample scale, particularly for rare classes (CRJ9, DH8D) with only 50 training segments.

7.3 Why the Transformer Underperforms

The Transformer drops to 63.5 % on the full dataset, below the sample TCN, despite 500x more training data. Likely causes:

- Short sequences (~81 timesteps): self-attention provides limited benefit over local convolutions at this scale.
- Strong positional structure: climb profiles vary strongly by phase; the implicit locality of dilated convolutions may encode this more naturally than learned positional embeddings.
- Hyperparameter sensitivity: a larger d_model, more layers, or longer training may be needed.

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8 Conclusion

The full-dataset TCN achieves 81.8 % accuracy (macro F1 = 0.755) on 11-class aircraft type classification from three-feature climb trajectories. Adding more data helps more than improving the architecture. The Transformer is not competitive on this short-sequence, locally structured task without further tuning.

Immediate next steps:

1. Feature expansion: add groundspeed or derived energy-height signals to break A319/A320/A321 degeneracy.
2. Per-class error analysis: visualise representative misclassified trajectories.
3. TCN hyperparameter search: more dilation levels, wider channels, larger kernel.
4. Ensemble: average TCN + CNN logits (+2-5 pp at low cost).
5. Confidence calibration: temperature scaling for operational use.

9 References

Alligier, Richard. 2017. "Climbing Aircraft Dataset." ENAC (École Nationale de l'Aviation Civile); OpenSky Network. https://opensky-network.org/datasets/#publication-data/climbing-aircraft-dataset/.

Alligier, Richard, and David Gianazza. 2018. "Learning Aircraft Operational Factors to Improve Aircraft Climb Prediction: A Large Scale Multi-Airport Study." *Transportation Research Part C: Emerging Technologies* 96 (November): 72–95. https://doi.org/10.1016/j.trc.2018.08.012.